

Case Study: BP Deepwater Horizon, Part 1

At about 9:00 p.m. on April 20, 2010, the Deepwater Horizon drilling rig in the Gulf of Mexico was nearing the end of a difficult operation at the Macondo well. The crew had already dealt with technical surprises, design changes, and delay. Now they faced a crucial question: how should they interpret a negative pressure test, a test meant to show whether the well's barriers would hold when pressure was reduced? The public record later assembled by investigators and the court makes one formal point unusually clear. BP, as operator of the well, held final authority over the key judgment about whether that test counted as successful and whether operations would continue.

This case uses that final sequence to examine formal organization. Students are asked to look mainly at BP's formal organization, its decision structure, its procedures, its internal layers of authority, and its accountability system. The contractors matter in this case mainly because they provided inputs into BP's formal control system.

A short guide to drilling terms used in this case

The case uses a small set of technical terms. You do not need prior drilling knowledge. The terms below are only meant to make the organizational analysis easier.

Core well and equipment terms

Term	Plain meaning for this case
Operator	The company with formal responsibility for the well. Here, BP was the operator.
Rig	The offshore drilling platform and the crewed workplace from which drilling operations were carried out. Here, that rig was the Deepwater Horizon.
Wellbore	The drilled hole that leads down into the well.
Casing	Steel pipe placed inside the wellbore to stabilize the well.
Centralizers	Small devices attached to casing to help keep it centered in the wellbore.
Blowout preventer	A large emergency device on the seafloor designed to help seal the well if control is lost.

Core warning, testing, and sealing terms

Term	Plain meaning for this case
Cement slurry	The liquid cement mixture pumped into the well to help seal spaces around the casing.
Channeling	A condition in which gaps or paths can remain in the cement, allowing gas to move where it should not.
Kick	An unwanted inflow of oil or gas into the well. It is a warning sign that well control may be weakening.
Negative pressure test	A test used to check whether the well's barriers hold when pressure is lowered.
Cement bond log	A diagnostic logging test that can provide additional information about cement placement.
Temporary abandonment	A stage in which the well is secured before the rig leaves the site.

BP as the formal center of the project

The Macondo operation was not carried out by BP alone. Transocean owned the rig and supplied the drilling crew. Halliburton provided cementing services and mudlogging, meaning the monitoring of drilling returns and related data. Schlumberger personnel were available for a cement bond log earlier on April 20. Yet within this arrangement, BP occupied the formal center. Public court findings and regulatory reports indicate that BP, as operator, remained responsible for the well and for key judgments about well integrity, meaning whether the well was safely sealed and controlled.

That formal position mattered because authority and expertise were not distributed in the same way. Specialized knowledge was spread across several organizations, but final authority over the well remained concentrated with BP. On the rig, BP's Well Site Leaders functioned as the operator's representatives. The court record states that, although Transocean personnel had stop work authority, the BP Well Site Leader had the ultimate decision on whether the negative pressure test would be treated as successful. For an organization course, this is a strong example of hierarchy, decision rights, and accountability residing in one place even when many actors are involved in the discussion.

The same pattern appeared in BP's relationship with Halliburton. Court findings indicate that Halliburton's recommendations on cement design and execution were contractually framed as opinions only. The same record cites a BP internal standard stating that reliance on service company expertise was unacceptable. In plain organizational terms, BP designed a system in which outside experts could advise and perform technical work after approval, but BP itself retained the duty to review, approve, and ultimately stand behind the decision.

This arrangement gave BP a clear formal role. It also created a demanding organizational task. BP had to take advice produced partly outside its own boundaries, judge its adequacy, and make a final choice under time pressure. In this first case, that problem is treated mainly as a problem of internal organization inside BP: how much authority was concentrated, how advice was routed, and how internal control was supposed to work.

Exhibit 1. BP's formal position in selected decisions

Decision area	Who supplied input	What BP's formal role was
Well integrity judgments on the rig	Transocean drilling personnel, BP offshore staff, and at times BP onshore engineers	BP Well Site Leaders held final authority over whether key well conditions counted as acceptable for continuing work.
Cement design and execution	Halliburton technical recommendations and services	BP retained responsibility for reviewing whether the cement design was appropriate before approving execution.
Choice to use or not use a cement bond log	Schlumberger could provide the diagnostic service	BP decided whether to use the additional diagnostic option.
Management of plan changes	Internal BP planning and contractor-related inputs	BP procedures called for formal evaluation and approval of important changes.

Formal procedures were present, but their force was uneven

The public record does not describe Macondo as an operation without rules. Investigators reported that BP had Management of Change requirements and other formal control expectations. These procedures were meant to structure how important deviations from plan would be evaluated, documented, and approved. This matters because the project did not unfold as a stable execution of an original plan. Geological conditions changed. The drilling plan changed. The operation moved toward temporary abandonment while key assumptions were still being revised.

One documented example concerned the debate over well design in April 2010. BP personnel considered a long string design, in which one long production casing string would be run to the bottom, and a liner tieback option, which would have involved a different casing configuration. The public record indicates that the liner option was associated with an additional cost estimate in the range of \$7 million to \$10 million. The safest organizational reading is not that cost alone determined the decision. Rather, the design choice took place in a formal setting where technical judgment, procedural review, and economic consequences were all present at once.

Another example involved centralizers. Modeling suggested lower channeling risk with a larger number of centralizers, but only six centralizers with integrated stop collars were ultimately used. In this case, the technical detail matters less than the organizational structure around it. BP retained the decision right. Yet the decision depended on whether materials could be obtained, delivered, and made ready in time. Formal authority remained centralized, but the conditions needed to exercise that authority cleanly were imperfect.

A third clue came earlier, on March 8, 2010, when the well experienced a significant kick. According to the dossier, the BOEMRE report later criticized the absence of a formally adequate investigation of that event. For students of organization, this is not simply a drilling detail. It raises the question whether formal learning routines, such as reporting, escalation, and investigation, operated with enough force before the final failure sequence.

Exhibit 2. Selected formal control points before the blowout

Date or period	Control point	Why it matters for formal organization
March 8, 2010	Significant kick occurred	Tests whether reporting and organizational learning routines were activated after a major warning sign.
April 2010	Long string versus liner tieback decision	Shows how formal decision arenas combined technical judgment, procedures, and economic visibility.
April 14 to 17, 2010	Centralizer shortage and decision to proceed with six centralizers	Illustrates centralized authority operating under imperfect material readiness.
April 19, 2010	Final compressive-strength analysis for the cement job was still incomplete when BP proceeded	Shows a formal choice about proceeding while some relevant information remained unfinished.
April 20, 2010	Decision not to run a cement bond log although Schlumberger was available	Highlights a documented choice not to use an additional diagnostic mechanism.
April 20, 2010	Negative pressure test treated as successful despite anomalies	Concentrates hierarchy, test interpretation, and escalation in one decisive moment.

Onshore support and offshore authority inside the same formal structure

Macondo also shows that BP's formal organization was layered across locations. The dossier indicates that BP engineers in Houston were available to support the offshore operation. The public record documents communication between an onshore BP engineer and the offshore BP Well Site Leader about anomalies in the negative pressure test. Yet the same record suggests that the anomaly was not escalated in a way that altered the course of action.

This arrangement can be read in two ways. On one hand, onshore support extended BP's formal control capacity by adding expertise beyond the rig. On the other hand, the structure could still leave the decisive judgment with the offshore role in the moment of action. That makes Macondo useful for analyzing not only who had authority, but also how reporting lines work when one organizational layer can advise while another must decide quickly.

The key point in this case is internal to BP's formal organization. The problem was not simply that information existed outside BP. Rather, BP's own structure had to connect offshore authority, onshore expertise, procedures for escalation, and a high-pressure operational clock. When those elements did not reinforce one another, more formal layering did not automatically produce more control.

Accountability, budget, and schedule

The public record gives this case another unusually clear formal feature: visible performance parameters. By the spring of 2010, the project was significantly behind the schedule BP had communicated to the regulator. By the time of the blowout, the project was also well over budget, with a documented overrun of \$58.34 million. Investigators further recorded that project personnel knew about the overrun and were concerned about avoiding additional unnecessary costs.

Students should use this material carefully. The evidence does not justify a simple psychological claim that cost pressure caused the disaster. But the record is strong enough to show that BP's formal accountability system made cost and schedule highly visible. In one strand of the record, cost containment appears to have been part of how performance was evaluated within a key well leadership role. In organizational terms, that means safety-related choices were being made inside a formal setting in which delay and extra expense were not neutral background conditions.

Formal accountability systems do not need to dictate a decision to matter. They can change what becomes salient, what counts as delay, and what appears to require justification. The case study therefore asks students to analyze how BP's formal organization, its hierarchy, procedures, reporting structure, and accountability pressures, may have shaped action before the blowout.

Exhibit 3. Formal features to examine in this case

Formal feature	What the record documents	Question for analysis
Hierarchy	BP Well Site Leaders held final authority over key well judgments on the rig, even though Transocean personnel had stop work authority.	How do clear decision rights work when advice, objection, and final authority are separated?
Procedures	BP had Management of Change requirements and other formal control expectations.	What happens when important procedures exist but important changes or anomalies are not backed by adequate evaluation?
Reporting lines	BP onshore expertise was available, but the negative-test anomaly was not escalated in a way that changed the outcome.	When does layered support strengthen control, and when can it weaken practical escalation?
Decision rights over diagnostics	BP decided not to use the cement bond log even though the service was available.	How should organizations decide when to delay for more information?
Accountability systems	The project was behind schedule and over budget, and cost containment was visible in the record.	How can performance parameters shape attention without mechanically determining action?

Closing

By the evening of April 20, 2010, the Macondo operation had become an organizational test as much as a technical one. BP's formal organization concentrated authority, relied on procedures, layered onshore support over offshore execution, and made cost and schedule visible within the project. These arrangements were designed to control risk. The challenge for students is to examine how the same arrangements may also have made renewed diagnosis, stronger escalation, or interruption of work more difficult at the moment when control was most needed.

Eleven people were killed in the explosion and fire that followed, seventeen were injured, and the Deepwater Horizon sank two days later. Oil flowed from the well for eighty seven days before it was capped. How did BP's formal organization shape what key actors saw, decided, escalated, or failed to stop?

Orienting Questions for Students

1. Which elements of BP's formal organization were most important at Macondo, and how do you know from the case?
2. Where does the case show that formal control mechanisms were supposed to reduce uncertainty, and where do those mechanisms appear weak, incomplete, or not fully activated?
3. How would you explain the contribution of BP's formal organization to catastrophic failure without reducing the case to either individual error or one single bad decision?

Sources Used for This Case

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